The Liouville skeleton of a unimodular acyclic cluster model

Definitions, target statement, and Lean correspondence

Cluster skeleton formalization project

8 November 2026

Abstract

This document specifies the geometric theorem formalized by this repository. It separates the input data and newly defined predicates from the theorems which construct their witnesses. The accompanying proof blueprint gives the mathematical argument. The scope is a smooth real Liouville structure with complete flow and an explicit full dynamical skeleton.

1 Input data and the underlying space

Let $d \geq 0$ and let $B = (B_{ij}) \in M_d(\mathbb{Z})$ be skew-symmetric with $\det B = 1$. Choose a set $M \subseteq \{1, \dots, d\}$ of mutable indices; write F for its complement. The directed graph on M, with an arrow $i \to j$ when $B_{ij} > 0$, is required to be acyclic. There is no acyclicity assumption on the full graph. The determinant condition concerns the entire matrix, including frozen indices.

For $a \in \mathbb{C}^d$ put

$$P_i(a) = \prod_{j=1}^{d} a_j^{[B_{ij}]_+}, \qquad Q_i(a) = \prod_{j=1}^{d} a_j^{[-B_{ij}]_+}, \qquad [n]_+ = \max(n, 0).$$

Define the actual space of complex exchange points

$$X_B = \left\{ (a, b) \in \mathbb{C}^d \times \mathbb{C}^M : \; a_i b_i = P_i(a) + Q_i(a) \; (i \in M), \quad a_f \neq 0 \; (f \in F) \right\}.$$

It has the subspace topology from the indicated vector space. The frozen coordinates are variable points of $\mathbb{C}^*$, not specialized coefficients. The point $a_j = 1$, $b_i = 2$ shows that this space is nonempty. Its initial torus is $T = \{a_1 \cdots a_d \neq 0\} \subseteq X_B$.

Lean correspondence: `ClusterSkeleton/Seed.lean`; `Seed`, `Seed.Unimodular`, `Seed.Point`.

2 Two independent subsets

Definition 2.1 (Coordinate cap)**.** The coordinate cap is the subset

$$C_B = \left\{ (a, b) \in X_B : \; |a_f| = 1 \; (f \in F), \quad |a_i| \leq 1, \; (1 - |a_i|^2) b_i = 0 \; (i \in M) \right\}.$$

In particular, $|a_i| < 1$ forces $b_i = 0$. At $|a_i| = 1$ the product condition imposes no further restriction on b_i; the exchange equation still applies. This definition contains no vector field or flow. Compactness of C_B is a theorem, not a field supplied with the definition.

Lean correspondence: `ClusterSkeleton/CapSet.lean`; `coordinateCap, isCompact_coordinateCap`.

Definition 2.2 (Reference strata)**.** Call $I \subseteq M$ *independent* if $B_{ij} = 0$ for all $i, j \in I$, and for $I \subseteq \{1, \dots, d\}$ put

$$\Lambda_I = \{(a, b) \in X_B : \ b_i = 0, \ |a_i| \le 1 \ (i \in M \cap I), \quad |a_j| = 1 \ (j \notin I)\}.$$

Each Λ_I is closed, and compact when $I \subseteq M$. For independent I this is the reference's stratum. The cap is their union,

$$C_B = \bigcup_{I \subseteq M \text{ independent}} \Lambda_I,$$

because a cap point lies in the stratum indexed by its own set of strict disk directions $\{i \in M : |a_i| < 1\}$, which is independent, while every Λ_I with $I \subseteq M$ lies in C_B. Independence is needed for neither inclusion; it only indexes the union by the reference's family.

Lean correspondence: `ClusterSkeleton/CapStrata.lean`; `lambdaStratum, lambdaStratum_subset_coordinateCap, coordinateCap_subset_lambdaStratum_strictCapIndices, coordinateCap_eq_iUnion_lambdaStratum`.

Definition 2.3 (Dynamical skeleton convention)**.** For a continuous real flow Φ on X_B, let

$$\mathrm{Skel}(\Phi) = \left\{ x \in X_B : \overline{\{\Phi_t(x) : t \ge 0\}}^{X_B} \text{ is compact} \right\}.$$

The closure is taken in the actual space X_B. This is the *relatively compact forward orbit* convention. It does not assert that points are fixed, and it does not use C_B in its definition.

Lean correspondence: `ClusterSkeleton/TrappedSet.lean`; `trappedSet`.

3 The geometric theorem

Theorem 3.1 (Formalized target)**.** *For every input (d, B, M) above, there exist a smooth real manifold structure on the fixed topological space X_B of real dimension $2d$, a smooth real one-form λ, a smooth vector field Z, and a jointly smooth flow $\Phi : \mathbb{R} \times X_B \to X_B$, with the following properties.*

1. *The actual inclusion $X_B \hookrightarrow \mathbb{C}^d \times \mathbb{C}^M$ is a smooth immersion, and T is dense in X_B.*
2. *$\omega = \mathrm{d}\lambda$ is nondegenerate, and on T it satisfies*

$$\omega = \mathrm{Im}\left(\frac{1}{2} \sum_{i,j=1}^{d} B_{ij} \, \mathrm{d}\log a_i \wedge \mathrm{d}\log a_j \right).$$

   The differential $\mathrm{d}\log a_i$ means $\mathrm{d}a_i/a_i$; no global logarithm is chosen. Since $\omega = \mathrm{d}\lambda$, the form is closed.

3. *$\iota_Z \omega = \lambda$. For every $x \in X_B$ and $t \in \mathbb{R}$,*

$$\Phi_0(x) = x, \qquad \Phi_{t+s}(x) = \Phi_t(\Phi_s(x)), \qquad \frac{\mathrm{d}}{\mathrm{d}t}\Phi_t(x) = Z(\Phi_t(x)).$$

   Thus this same vector field is complete in both time directions.

4. *For this same flow, there is literal equality of subsets of X_B:*

$$\boxed{\mathrm{Skel}(\Phi) = C_B = \bigcup_{I \subseteq M \text{ independent}} \Lambda_I.}$$

The second equality is the reference's stratified form of the identity. It is a restatement of the same proved equality through the cap description above, not a further claim.

Lean correspondence: `ClusterSkeleton/FullSkeleton.lean`; `trappedSet_eq_coordinateCap, trappedSet_eq_iUnion_lambdaStratu`

The quantifier order is

$$\forall (d, B, M) \text{ as above}, \quad \exists (\text{smooth structure}, \lambda, Z, \Phi) \quad \text{with all four properties.}$$

Neither a flow, completeness, compactness of the cap, nor either skeleton inclusion is an additional hypothesis. The construction uses positive cutoff parameters ε, δ and $\kappa \geq 2\varepsilon + 1$. The final existence theorem chooses $\varepsilon = \delta = 1$, $\kappa = 3$; these are auxiliary parameters, not values of frozen coordinates.

4 Definitions, formulation, and proof in Lean

Role	Declaration and meaning
Input definition	`Seed`, with acyclicity as input; `Seed.Unimodular`, the separate determinant hypothesis.
Space and sets	`Seed.Point`, `coordinateCap`, and `trappedSet` define the objects being compared; `lambdaStratum` defines the strata Λ_I.
Witness interface	`SmoothPresentation` and `LiouvilleData` specify the required constructions. Defining these structures does not construct an inhabitant.
Target proposition	`LiouvilleSkeletonTarget` and `FirstGeometricTarget` state the quantified goal.
Constructed witnesses	`SmoothAtlas.presentation` and `FullSkeleton.liouvilleData` supply the actual geometry.
Equality theorem	`FullSkeleton.trappedSet_eq_coordinateCap` combines the two proved inclusions for the same flow; `FullSkeleton.trappedSet_eq_iUnion_lambdaStratum` restates it as the union of the strata, via `coordinateCap_eq_iUnion_lambdaStratum`.
Chart rank	`IndependentNodeChart.complementRank_eq_sub_two_mul_card` proves the node-chart torus rank is $d - 2\lvert I\rvert$ for independent I, with `IndependentNodeChart.two_mul_card_le` for $2\lvert I\rvert \leq d$.
Existence theorem	`FullSkeleton.firstGeometricTarget` proves `FirstGeometricTarget`.

All names in the last four rows lie in the namespace `ClusterSkeleton`. The core theorem is in `ClusterSkeleton/FullSkeleton.lean`; the target proposition is in `ClusterSkeleton/GeometryTarget.lean`.

Mathlib supplies native topology, manifolds, derivatives and flows. The pinned `fs-fltz-minimal` dependency supplies smooth-form and Liouville interfaces built on these notions. This project defines the specific cluster model and proves the required constructions and equations. New formal definitions are an encoding choice; they are not in themselves a claim of new mathematical concepts or a proof of the target.

5 Scope and verification

This is the Liouville/skeleton part of the original acyclic-cluster geometric problem. The statement does not include a holomorphic extension theorem for the complex logarithmic two-form, finite type, a product description of individual cap pieces, generalized cocores, or any Fukaya, microlocal, or homological mirror symmetry equivalence.

The repository's `verification/README.md` distinguishes historical Horizon source-bound acceptance from the publication checkout's reproducible checks. The Lean kernel checks the supplied proofs; a separate statement review checks that the definitions and quantifiers express Theorem 3.1. The accepted foundation allowlist is `propext`, `Classical.choice`, and `Quot.sound`; these are standard logical foundations, not additional geometric assumptions.

6 Updated reference and continuation scope

The accompanying proof blueprint now follows the supplied appendix "Skeleton in the acyclic case". It uses column exchange exponents and an integral unimodular completion E with a positive rational diagonal $\widetilde{D}$ such that $E\widetilde{D}$ is skew-symmetric. The existing formalized theorem above is the specialization $\widetilde{D} = I$. Row and column exchange conventions interchange the two monomials in this specialization; their sum is unchanged. The continuation uses the existing code to check the supplied argument in the requested skew-symmetric case. The weighted notation in the reference does not certify a general skew-symmetrizable Lean theorem. Unimodularity and mutable acyclicity remain hypotheses.